

ECE 5200: Introduction to Digital Signal Processing Spring 2026

Instructor: Kiryung Lee, [760 Dreese Lab](#), lee.8763@osu.edu

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E-Access: <http://carmen.osu.edu>

Lecture Hours: MWF 12:40PM–1:35PM, [Fontana Lab 2020](#)

Office Hours: TBD

Objective: The objective of this course is to provide an understanding of the relationships between continuous-time signals and their sampled versions, the relationships between various notions of frequency domain, and signal processing techniques applied to discrete-time signals, especially frequency-selective filters. Advanced topics and applications will also be discussed.

Prerequisites: • Signals & Systems (ECE 3050 or equivalent)
• Probability & Statistics (STAT 3470 or equivalent)

Text: • Lecture packets available at [CarmenCanvas](#).
• A.V. Oppenheim and R.W. Schaffer, *Discrete-Time Signal Processing*, 3rd edition, Pearson, 2010. (Any addition of the text, or similar text, will suffice. Available free for two hour reserve at the 18th Ave Library.)
• P.P. Vaidyanathan, *Multirate Systems and Filter Banks*, Prentice Hall, 1993. Available online at <https://authors.library.caltech.edu/109485/>
• P. Prandoni and M. Vetterli, *Signal Processing for Communications*. Available online at <https://www.sp4comm.org/download.html>

References: • S. Mallat, *A Wavelet Tour of Signal Processing: The Sparse Way*, 3rd edition, Academic Press, 2009. Available online at <https://www.sciencedirect.com/book/monograph/9780123743701/a-wavelet-tour-of-signal-processing>

Evaluation:	Homework (lowest score will be dropped)	30%
	Mini Projects (including Matlab coding)	20%
	Midterm exams (in-class February 20 and April 3)	30%
	Final exam (12:00PM–1:45PM, Tuesday May 5)	20%

Homework: Homework problems will be assigned and collected on a regular basis; certain of these problems will be graded. Homework assignments and due dates will be posted at Carmen. Please submit a scan or photocopy of homework report to [CarmenCanvas](#). The lowest score will be dropped. **A homework score above 65.0% is required to pass the course.** Discussions on homework are encouraged. However, all written work turned in must be your own. All sources and collaborators must be properly acknowledged. For instance, if students are involved in the collaboration, please include a list of their names.

Readings: Suggested reading assignments for specific sections in the text will be listed on the homework assignment sheets.

Exams: The examinations will be closed-book with a single one-sided $8'' \times 11''$ sheet of notes allowed for each midterm exam, and three sheets (one-sided) for the final exam. Synthesizing material into an organized note sheet is a good exercise for learning and retention. Students seeking to reschedule an exam due to religious observance or holiday should notify the instructor at least ten days prior to the scheduled exam date. In cases of medical or family emergencies, a conflict exam may be requested with appropriate written documentation verifying the circumstances.

Special Needs: Any student who feels they may need an accommodation based on the impact of a disability should contact the instructor privately to discuss specific needs. Please contact the OSU Office for Disability Services for assistance in verifying the need for accommodations and developing accommodation strategies.

Letter Grade Distribution:

		A	95–100	A-	91–94.99
B+	87–90.99	B	83–86.99	B-	79–82.99
C+	75–78.99	C	71–74.99	C-	67–70.99
D+	63–66.99	D	59–62.99	D-	55–58.99

Disabilities Statement

Any student who feels s/he may need an accommodation based on the impact of a disability should contact the instructor privately to discuss specific needs. Please contact the OSU Office for Disability Services for assistance in verifying the need for accommodations and developing accommodation strategies. If you require accommodation, please let the instructor know at the beginning of the term, and not later than the second Friday, so that if arrangements are needed regarding midterms, there will be time to set them up.

Academic Misconduct Statement

Any student found to have engaged in academic misconduct, as set forth in the Code of Student Conduct Section 3335-23-04, Prohibited Conduct, will be subject to disciplinary action by the university. Academic misconduct is any activity that tends to compromise the academic integrity of the university, or subvert the educational process.

Student Conduct

Students are expected to abide by the provisions in the Code of Student Conduct. The University's Code of Student Conduct and Sexual Misconduct Policy are available on the OSU Web page.

Topical Outline

introduction

signals & systems review

sampling and conversion

processing of discrete-time signals

design of discrete-time filters

multirate systems

sub-Nyquist sampling

time-frequency analysis

sparse signal processing and inverse problems

The instructor reserves the right to make any changes he considers academically advisable. Such changes, if any, will be announced in class and at **CarmenCanvas**.